
Data Science/ Analytics Intern – Round-0 Assignment (Trader Performance vs Market Sentiment)

Hello

Thanks for applying for the **Data Science Intern** role at **Primetrade.ai**.

As the first step in the hiring process, please complete the short assignment below. Candidates who meet the acceptance criteria will be shortlisted for an interview.

Expected effort: ~2–3 hours

Deadline: Submit within **2 days** of receiving this email (*If you need 1–2 extra days, reply with your ETA.*)

Objective

Analyze how **market sentiment (Fear/Greed)** relates to **trader behavior and performance** on Hyperliquid. Your goal is to uncover patterns that could inform smarter trading strategies.

Datasets

You will work with 2 datasets:

1) Bitcoin Market Sentiment (Fear/Greed)

Columns: **Date**, **Classification** (Fear / Greed)

Link: https://drive.google.com/file/d/1PgQC0tO8XN-wqkNyghWc_-mnrYv_nhSf/view?usp=sharing

2) Historical Trader Data (Hyperliquid)

Includes fields like: **account**, **symbol**, **execution price**, **size**, **side**, **time**, **start position**, **event**, **closedPnL**, **leverage**, etc.

Link: <https://drive.google.com/file/d/1IAfLZwu6rJzyWKgBToqwSmmVYU6VbjVs/view?usp=sharing>

Please check for the download symbol in the left link

Tasks (What to do)

Part A — Data preparation (must-have)

1. Load both datasets and document:
 - number of rows/columns
 - missing values / duplicates
2. Convert timestamps and align the datasets by **date** (daily level is fine).
3. Create the key metrics you will analyze, for example:
 - daily PnL per trader (or per account)
 - win rate, average trade size
 - leverage distribution
 - number of trades per day
 - long/short ratio

Part B — Analysis (must-have)

Answer these questions with evidence:

1. Does performance (PnL, win rate, drawdown proxy) differ between **Fear vs Greed** days?
2. Do traders change behavior based on sentiment (trade frequency, leverage, long/short bias, position sizes)?
3. Identify 2–3 segments (examples):
 - high leverage vs low leverage traders
 - frequent vs infrequent traders
 - consistent winners vs inconsistent traders
4. Provide at least **3 insights** backed by charts/tables.

Part C — “Actionable output” (must-have)

Propose **2 strategy ideas** or “rules of thumb” based on your findings.

Example: “During Fear days, reduce leverage for segment X; increase trade frequency only for segment Y.”

Bonus (optional)

- Simple predictive model:
 - Predict next-day trader profitability bucket or volatility of PnL using sentiment + behavior features
 - Clustering traders into behavioral archetypes
 - A lightweight dashboard (Streamlit) to explore results
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Deliverables (What to submit)

Please submit **one** of the following:

Option 1 (Preferred): GitHub repo

Include:

- Notebook (`.ipynb`) or script
- `README.md` with setup + how to run
- Output charts/tables
- A short write-up (max 1 page) OR a markdown section summarizing:
 - methodology
 - insights
 - strategy recommendations

Option 2: Google Drive folder

Same contents as above.

Evaluation criteria (How we shortlist)

We score based on:

- Data cleaning + correctness of merges/alignment
- Strength of reasoning (not just plots)
- Quality of insights (actionable, not generic)
- Clarity of communication (structured write-up)
- Reproducibility (clean notebook, clear steps)

How to Apply

PLEASE SUBMIT THROUGH GOOGLE FORM DOC SHARED NOT BY EMAIL

We'll notify shortlisted candidates by Saturday after submission.

Thanks,
Sonika
Primetrade.ai Hiring Team
